

Data Representation

Machine Language

[*] what is **0b01100**

Bits & Bytes

bit = 0 or 1
nibble = 0000, 1111
byte = 00000000, 11111111

Units:

short (word) = 00000000 00000000
int (dword) = 00000000 00000000 00000000 00000000
long (qword) = 00000000 00000000 ... 00000000 00000000

** word = register size*

Different Bases

$$\begin{aligned} \mathbf{X}_{10} &= 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 \\ \mathbf{X}_2 &= 0, 1 \end{aligned}$$

$$\mathbf{X}_{16} = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F$$

$$\begin{aligned} 19304779100973290_{10} &= \mathbf{X}_2 \quad ??? \\ 11000100110010101_2 &= \mathbf{X}_{10} \quad ??? \end{aligned}$$

$$\begin{aligned} \text{D5C4}_{16} &= 1101010111000100_2 \\ 1101001111101001_2 &= \text{D3E9}_{16} \end{aligned}$$

Characters

...

A = 65 = 0x41

B = 66 = 0x42

C = 67 = 0x43

...

"AAAA" = 41 41 41 41

"0123" = 30 31 32 33

* ASCII characters, 1 byte long

Signed vs Unsigned

```
int x = 1;
unsigned int y = 1;

int z = 4294967295;
unsigned int w = 4294967295;
```

dump

...	00	00	00	01	...
...	00	00	00	01	...
...	FF	FF	FF	FF	...
...	FF	FF	FF	FF	...

Integer Overflow

```
unsigned char x = 255;  
x++;
```

```
printf("%d", x);
```

```
> ./program  
0
```

Endianness

```
unsigned char x = 255;  
short        y = 0x1337;  
int          z = 0x00D34D00;  
unsigned     w = 0x12345678;  
  
char s[] = "0123";
```

dump (BE)

...	FF	...			
...	13	37	...		
...	00	D3	4D	00	...
...	12	34	56	78	...
...	30	31	32	33	...

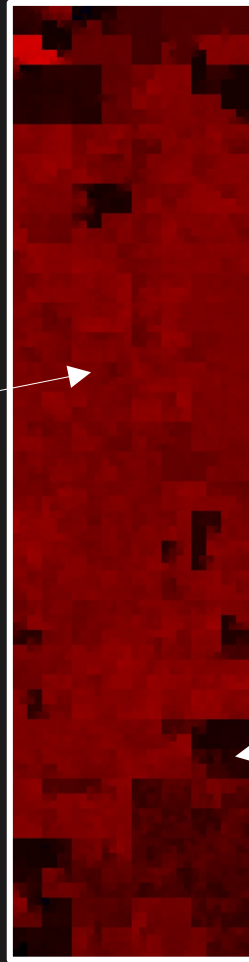
dump (LE)

...	FF	...			
...	37	13	...		
...	00	4D	D3	00	...
...	78	56	34	12	...
...	30	31	32	33	...

Entropy

```
[*] high = compression, encryption  
[*] low  = data, code
```

high entropy ($\rightarrow 1$)



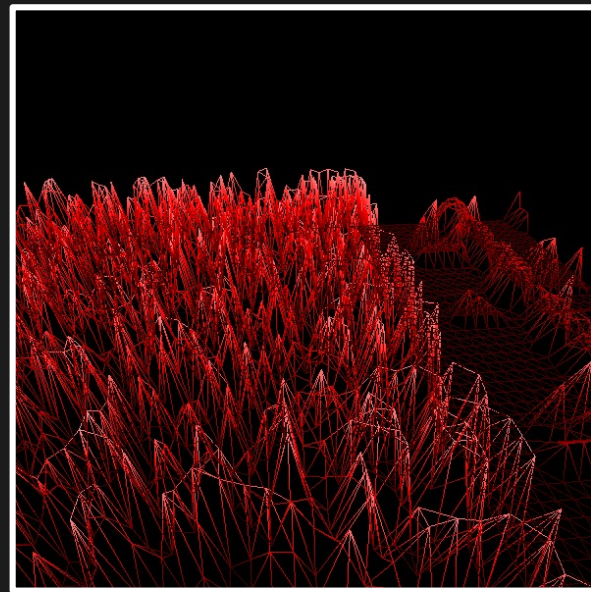
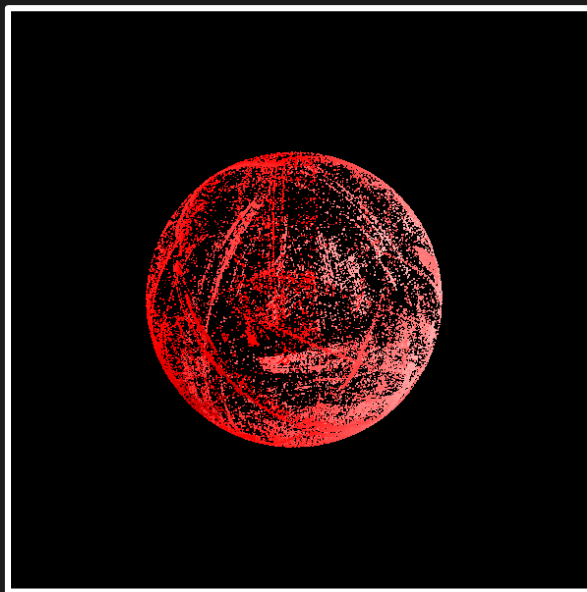
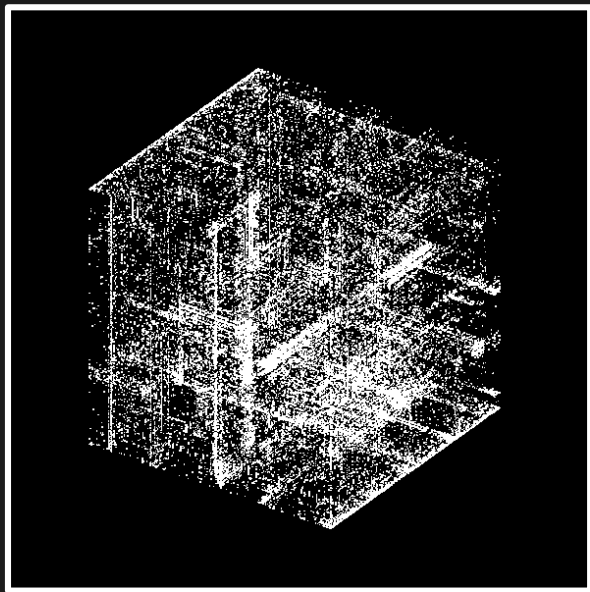
low entropy ($\rightarrow 0$)

Forensic

ImHex



Visual RE



Questions?